Apertus-Recht: which racial-discrimination law does Apertus apply?

People ask chat assistants whether something said at dinner, in a chat or at a rally is punishable. We release 1,980 German, French and Italian minimal-twin scenarios under Art. 261bis StGB. Each verdict is computed by code from pinned Fedlex texts and BGE 130 IV 111, and agrees with the 58 Swiss court rulings we could code on the modelled elements (55 convictions, 3 acquittals; a check of the rules' conjunction, not of how courts classify facts). A harness re-derives every label and number from a clean checkout. Asked without the law, Apertus v1.5 70B convicts every scenario we wrote (balanced accuracy 0.50). In free text it calls 73 of 75 scenarios punishable that are not punishable under Art. 261bis. With the verbatim Fedlex text and the Regeste of the leading case pasted in, it still convicts everything; only rules spelled out in plain sentences make it apply the publicity rule. It still convicts for a canton, a disability or a political opinion, and for acts committed before the law existed. One sentence on when sexual orientation became protected makes it stop convicting such cases in German and French.

Track: Apertus Readiness · Track 1B (Swiss Voices, behavioural alignment) · **Event:** Online · **Team:** 444
Demo: `make run` in the repository (Docker, CPU, no key, ~25 s); dataset on Hugging Face (link in the form notes)

Use of Apertus

- **Model:** Apertus v1.5 70B; Apertus v1.5 8B (public Q4_K_M GGUF; exploration phase: SFT, RLVR, DPO and final stages, bf16 weight blends, Q8_0 and MLX 4-bit builds).
- **How it is used:** evaluation (forced choice, preference pairs, free text with three blind LLM judges, two Claude and one Qwen).
- **Where it runs:** the 70B on Public AI via a cost-capped gateway (the endpoint name `apertus-v1.5-8b` serves the 70B: byte-identical output to `-70b` in 760/792 cells, `docs/builds.md`); the 8B GGUF on CPU in GitHub Actions.
- **Links:** code https://github.com/mi11ione/apertus-recht · dataset on Hugging Face (URL in the form notes).

Use Case Description

Art. 261bis StGB (racial discrimination; extended to sexual orientation on 1 July 2020) is litigated in every language region. Whether a statement is punishable turns on three elements a layperson cannot check: whether it was **public** (under BGE 130 IV 111 E. 5.2, statements among family, friends or in a setting of personal trust are not), whether the targeted characteristic is on the **closed list** (race, ethnicity, religion, sexual orientation), and **when** the act took place.

A wrong answer misinforms people about their own speech in either direction. The user community is Swiss residents asking a local assistant in German, French or Italian, with or without the law in its context. Apertus-v1.5-8B was downloaded 698,641 times in the 30 days before 9 October 2026. The article itself was adopted by referendum in 1994 (54.6% yes) and extended to sexual orientation by referendum in 2020 (63.1% yes). Its training data include `voilaj/swiss-caselaw`, which contains BGE 130 IV 111 verbatim, plus 147 decisions citing it. The dataset asks not "how often is the model right" but **which law it applies**, element by element.

Dataset Summary

Per language: 660 scenarios in 220 twin groups (220 base, 215 true flips, 210 placebos, 15 contested). Verdicts: 430 convict, 215 acquit, 15 open. Each scenario comes with an acceptance criterion, an item-specific `chosen` answer, a `rejected` answer of the kind models give and a length-matched mirror, a settled or contested label, the normative source and the element-by-element derivation; prompts at six rungs (R0 no law; R1–R4 the elements, a one-sentence paraphrase of BGE 130 IV 111, the closed list, the 2020 date; R5 the verbatim Fedlex text of Art. 261bis and Art. 2 with Fedlex's footnote, and the Regeste) in two forced-choice framings (Ja/Nein; "Urteil: schuldig / nicht schuldig"); a free-text question; and a preference prompt. The same 660 German scenarios are also told in a user's own words (`de_lay`).

Family (groups per language)	One-fact change of the base	Tests
SET (55)	public → private setting; placebo: family wording of a public statement	BGE 130 IV 111 E. 5.2
CHAR (60), CHARX (20)	protected → unprotected characteristic; placebo: synonym ("Hautfarbe", "Glaube")	closed list
DATE_SO (35)	sexual orientation 2021–25 → before 1 Jul 2020; placebo: 1 Jul / Aug 2020	AS 2020 1609, Art. 2 StGB
DATE_CTRL (25)	race, ethnicity, religion → 2018–20 (placebo) and → 1993/94	over-application; entry into force
CONTESTED (15)	public → fact-dependent setting (gold open)	hedging
WORDING (10)	another wording family; another paragraph	wording invariance

Unprotected characteristics: gender identity, sex, canton, occupation, age, disability and political opinion. Paragraphs are balanced (Abs. 1/2/4: 220/218/222 in German). The wording is deliberately templated: 18 act wordings in three families, 29 settings, 33 group phrases. A mean of 44 words is the price of exact twins.

What is new. To our knowledge no existing dataset lets one test which version of a real statute a model applies, element by element, in the three official languages (comparison with sources in `docs/prior-art.md`):

Dataset	Languages	Gold	One-fact twins	Real amendment dates, non-retroactivity	Checked against courts
LEXam / SwissLegalEvals	DE, EN / DE, FR, IT	exam keys, LLM judges	no	no	no
LawShift (NeurIPS 2025)	ZH	expert metamorphic relations	yes	hypothetical revisions only	no
TALD (2026)	ZH	statute version cited in judgments	no	which version applies	from judgments
OpenExempt (2026)	EN	solver over expert encodings	single-factor suites	one deadline rule	no
Multilingual HateCheck	DE, FR, IT, …	by construction	templated contrasts	no (not legal)	no
Apertus-Recht	**DE, FR, IT**	**code from pinned Fedlex texts and BGE**	**yes, with placebos**	**1 Jan 1995, 1 Jul 2020, Art. 2 StGB**	**58/58 (3 acquittals)**

Data Collection Method

No data was collected from people. A seeded generator renders the scenarios from YAML templates. A 97-line engine computes the gold verdict from a machine-readable element model of the statute (`statutes/stgb_261bis/statute.yaml`). Every element and value cites a pinned source: the Fedlex consolidations of Art. 261bis before and since 1 July 2020 and of Art. 2 StGB (DE/FR/IT, sha256 of the text), the amendment footnotes (AS 1994 2887, AS 2020 1609) and the Regeste of BGE 130 IV 111. The engine, not the author, labels each variant a true flip, a placebo or contested. The generator refuses to build if the label disagrees with the design.

Design choices. Placebos catch surface heuristics; rungs model an assistant's retrieval levels; two framings, since a framing is readable only if it convicts the protected reference; statute vocabulary, no slurs or names; 15 contested items per language leave BGE 130 IV 111's decisive facts mixed (gold open, hedged answer expected).

Translation and localisation. FR/IT by Claude agents with the official CP terms, blind back-translation (0 critical findings), then a lawyer-persona review that localised what back-translation cannot see (A7): Romandie and Ticino places, "avec conscience et volonté", "condamnation / acquittement", Swiss citation forms.

Verification without human labels. Gold is computed, so review looks for errors in the code, the element model or the wording. All review was done by Claude agents working blind on opaque ids, without access to code or gold.

- **Annotation:** six agents labelled 960 items. They agreed with the code on 939/939 settled items and on the deciding element in 315/315 acquittals. A second pass that gave the agents only the law (statute texts and Regeste, no rule summary) agreed on 939/939.
- **Wording fixes:** reviewers found two wording defects before any result was analysed (amendment A3) and four after (A5). All were fixed, the affected prompts re-scored, and the pre-fix responses archived.

- **Adjudication:** two independent adjudicators covered every disagreement, every contested item and a blind 10% sample. Gold was wrong in 0/94 sampled items (Wilson upper bound 3.9%).
- **Court validation:** 326 decisions citing Art. 261bis triaged, each extracted twice (κ 0.83–0.98, a third agent on conflicts): the engine agrees with 58/58 in-scope rulings, a strong test of the conviction rules and a thin one of the acquittal rules (an always-convict engine scores 55/58; 3 acquittals, all agree).

Dataset Details

Quality gates (`statute-twins qc`, all passing): schema; gold recomputed for every item; twin invariants (one changed slot; placebos keep the verdict, flips change it); 0 duplicates; language identity; canary; a PII scan over items, responses and metadata (0 hits). Tests check every closed-list term and both in-force dates against the official text in three languages. Mutation tests (drop religion from the list, move the 2020 date by a day, make private settings public) flip exactly the items each rule governs.

Contamination. None of our 35,241 distinct 13-grams occurs in the 1,069,365 decisions of `voilaj/swiss-caselaw`; the same code finds 846 of the 1,525 13-grams of the statute and the Regeste. The law itself is well represented: 147 decisions mention BGE 130 IV 111 and 754 mention Art. 261bis.

Ethics and licensing. No data subjects: actors are generic, groups are named by their characteristic, acts are abstract. Items are CDLA-Permissive-2.0 with per-item licence components; statute and decision texts are public domain (Art. 5 URG). AI use is declared throughout; a consent form with an AI-use clause is ready for anyone who adds human reviewers. Every item carries a canary string.

Evaluation

Three measurements, one item set.

- *Forced choice* (all items, R0/R3/R4 and, in German, R5; two framings): the summed log-probability of the convict continuation minus the acquit one. On vLLM this is exact (prefilled answer, prompt log-probabilities); in llama.cpp it is read from the logits.
- *Preference accuracy* (settled items, R0): the same exact scoring between each item's `chosen` answer and its length-matched mirror, the same sentence with the verdict and the deciding legal claim negated (median length ratio 1.0; a first design with a shorter `rejected` answer measured length and was discarded before analysis, A8).
- *Free text* (a fixed sample of 240 German items, R0 and R3, greedy, 220 tokens): every answer is judged against its acceptance criterion by two independent Claude judges, blind to model, rung and family.

The read-back pairs every true flip (variant V) with its base (reference Ref, which the statute convicts) and takes 95% item-bootstrap intervals of the mean log-odds of conviction. A framing is *readable* (anchored) for that pair set when the interval of Ref lies above zero, i.e. the model convicts the reference at all. An element is *misapplied* when V's interval also lies above zero (the model convicts where the law acquits), *applied* when it lies below. It is **red** when misapplied in every readable framing. The pre-registered test fits this read-back on wording families A and B and predicts family C; it passes if it beats the statute by at least 0.15 in every readable framing. R1 and R2 (the elements named; the publicity rule added) are released for finer retrieval studies; we score R0, R3 and R4.

Runs: the 70B on Public AI (vLLM, temperature 0) on all three languages at R0/R3/R4 and on German at R5 (13,200 decisions), plus the German user voice at R0; the 8B public Q4_K_M GGUF (sha256 `a037df8d…`) on CPU CI runners, German R0/R3/R4/R5 (5,280); preference on 645 settled items per language; free text on 480 German answers; the exploration battery (162 items) for training stages and builds. Every row records the served model and a request hash.

Results

Without the law, the 70B convicts everything (R0: recall of convictions 1.00, of acquittals 0.00, in every language and framing). It is not blind to the facts: its conviction scores still rank punishable above non-punishable cases (AUC 0.72–0.77), but the threshold sits so far on the convicting side that no case falls below it. Free text agrees with the forced choice. The judges failed 73 of 75 R0 answers to acquittal items as misleading, and every one of the 14 contested-item answers gave a flat verdict. The two Claude judges agree on 718 of 720 pass decisions (κ 0.993); a judge from another model family (Qwen-SEA-LION-v4-32B) agrees with their consensus on 694 of 718 (139 of 143 on the 20% sample the track asks to check), and a deterministic parser of each answer's verdict agrees with the judges' reading on 698 of 700 settled answers.

model	lang	R0 bal. acc.	R3 bal. acc.	R4 bal. acc.	R5 bal. acc.	R3 acquit recall
70B	DE	0.50 / 0.50	0.74 / 0.70	0.73 / 0.75	0.50 / 0.50	0.50 / 0.40
70B, user voice	DE	0.50 / –	0.74 / –	–	–	0.55 / –
70B	FR	0.50 / 0.50	0.76 / 0.83	0.81 / 0.77	0.53 / 0.50	0.56 / 0.71
70B	IT	0.50 / 0.50	0.71 / 0.70	0.84 / 0.75	0.60 / 0.58	0.45 / 0.40
8B Q4_K_M	DE	0.52 / 0.50	0.78 / 0.72	0.80 / 0.77	0.53 / 0.51	0.57 / 0.46

Table 1. Balanced accuracy (mean of the recall of convictions and of acquittals; 0.50 = no discrimination), Ja/Nein / verdict framing, and the recall of acquittals at R3. The user voice was run in German only, at R0 and R3.

With the rules spelled out, publicity is applied but the list and the date are not. Figure 1 shows the read-back. For the German 70B:

- **R3** (the elements, a one-sentence paraphrase of BGE 130 IV 111, the list and "the list is closed"): private settings are acquitted. Canton, disability and political opinion stay red, convicted although the prompt lists the four protected characteristics.
- **Gender identity** is applied once the prompt names it as excluded.
- **R4** adds the sentence "sexual orientation is protected since 1 July 2020". The 70B then acquits the cases before that date, but also stops convicting punishable cases from 2021–25. With the verdict word it convicts 0, 1 and 5 of 35 in German, French and Italian; with Ja/Nein, 3 (German), 11 (French) and 29 (Italian) of 35. The 8B convicts 13 of 35. The clause does not teach the date; it takes the characteristic off the list. Recall of convictions falls to 0.74 (grey cells in Figure 1).
- **Acts before 1995**, when the article did not exist, are convicted at every rung. No rung states the 1995 date, and Fedlex's own footnote omits it, so this is a gap in what the model knows.
- **R5** pastes the law itself, as a retrieval pipeline would: the Fedlex text of Art. 261bis and Art. 2 StGB with Fedlex's footnote, and the Regeste of BGE 130 IV 111. The 70B convicts everything again, exactly as with no law at all (acquittals recognised: 0 of 215 with Ja/Nein, 1 with the verdict word). The verbatim text enumerates the four characteristics and the Regeste states the publicity rule, yet they help less than our one-sentence paraphrases. R5 also lacks R3's instruction to check the elements; a control with that instruction added (R6, Ja/Nein) recognises 23 of 215 acquittals, against 107 at R3. In free text with the verbatim law, 75 of 75 answers to non-punishable scenarios say they are punishable. In French and Italian the verbatim law recognises 12 and 43 of 215 acquittals with Ja/Nein, against 120 and 97 with the rules spelled out.

Not an artefact of legal register. The same German scenarios told in a person's own words ("Im Januar 2024 habe ich in Aarau an einem Stand auf dem Wochenmarkt … ein Weltbild propagiert, nach dem Muslime minderwertig sind") give the same profile: 2 of 215 acquittals at R0; at R3 balanced accuracy 0.74 (charge-sheet form 0.74), with private settings acquitted, the closed list half applied and the dates not at all (Ja/Nein; the prompt still names the article).

Italian shows the same profile. In French the closed list holds better at R3 (canton 12/16, disability 6/9 convicted).

One fact changed	Law	70B · Deutsch				70B · Français				70B · Italiano				8B Q4_K_M · Deutsch			
		R0	R3	R4	R5	R0	R3	R4	R5	R0	R3	R4	R5	R0	R3	R4	R5
Statement only among family or friends	acquit	55/55	5/55	6/55	55/55	54/55	0/55	0/55	43/55	54/55	8/55	3/55	12/55	52/55	14/55	0/55	42/55
Group defined by canton	acquit	16/16	13/16	9/16	16/16	16/16	12/16	2/16	16/16	16/16	16/16	14/16	16/16	14/16	10/16	4/16	16/16
Group defined by occupation	acquit	16/16	10/16	6/16	16/16	16/16	7/16	0/16	16/16	16/16	9/16	0/16	16/16	16/16	5/16	0/16	16/16
Group defined by age	acquit	17/17	4/17	1/17	17/17	16/17	4/17	1/17	17/17	17/17	7/17	1/17	17/17	17/17	0/17	0/17	17/17
Group defined by gender identity	acquit	16/16	0/16	0/16	16/16	16/16	0/16	0/16	16/16	16/16	0/16	0/16	16/16	14/16	0/16	0/16	16/16
Group defined by sex	acquit	18/18	0/18	0/18	18/18	18/18	0/18	0/18	18/18	18/18	2/18	0/18	18/18	15/18	0/18	0/18	18/18
Group defined by disability	acquit	9/9	8/9	4/9	9/9	9/9	6/9	2/9	9/9	9/9	9/9	2/9	9/9	9/9	0/9	0/9	9/9
Group defined by political opinion	acquit	8/8	8/8	7/8	8/8	8/8	7/8	1/8	8/8	8/8	7/8	2/8	8/8	8/8	4/8	3/8	8/8
Sexual orientation, act before 1 Jul 2020	acquit	35/35	35/35	0/35	35/35	35/35	35/35	0/35	35/35	35/35	35/35	0/35	35/35	31/35	35/35	1/35	35/35
Any group, act before 1995	acquit	25/25	25/25	25/25	25/25	25/25	24/25	22/25	25/25	25/25	25/25	25/25	25/25	24/25	25/25	13/25	25/25
Public statement in family wording	convict	70/70	60/70	47/70	70/70	70/70	52/70	38/70	70/70	70/70	59/70	51/70	69/70	68/70	66/70	41/70	70/70
Listed characteristic, other word	convict	60/60	60/60	52/60	60/60	60/60	60/60	50/60	60/60	60/60	60/60	51/60	60/60	59/60	60/60	46/60	60/60
Race, ethnicity, religion, act 2018–20	convict	25/25	25/25	25/25	25/25	25/25	25/25	24/25	25/25	25/25	25/25	25/25	25/25	24/25	25/25	17/25	25/25
Sexual orientation, 1 Jul or Aug 2020	convict	35/35	35/35	12/35	35/35	35/35	35/35	20/35	35/35	35/35	35/35	31/35	35/35	31/35	35/35	22/35	35/35

Figure 1. Synopse: the law in force (column "Law") next to the law Apertus applies. Each row changes one legal fact of a scenario the statute punishes (above the rule) or a detail that changes nothing (below). Cells: variants convicted (k/n, Ja/Nein framing). Red: the model applies a law that is not in force, in both framings; light red: in one; grey: the unchanged reference is not convicted either, so the element cannot be read.

measure	model	lang	all	convict	private	not listed	SO < 2020	< 1995
preference	70B	DE	0.98	0.98	1.00	1.00	0.97	1.00
preference	70B	FR	0.69	0.53	1.00	1.00	0.97	1.00
preference	70B	IT	0.94	0.92	1.00	0.98	1.00	1.00
preference	8B Q4_K_M	DE	0.82	0.82	1.00	1.00	0.00	1.00
free text R0	70B	DE	156/240	154/158	1/19	1/34	0/13	0/9
free text R3	70B	DE	187/240	150/158	19/19	17/34	0/13	0/9
free text R5	70B	DE	159/240	158/158	0/19	0/34	0/13	0/9

Table 2. Preference accuracy: share of settled items where the model gives the item's chosen *answer a higher log-probability than its length-matched mirror. Free text: answers that meet the acceptance criterion (both judges agree; 240 German items). Columns: the element that decides the item.*

Recognising the law is not applying it. Shown an item's correct answer and its mirror, the 70B picks the correct one (German share 0.98), most cleanly in the closed-list and date columns, where both answers make a legal claim of the same form, although its own verdicts convict those cases. Elsewhere the mirror contradicts the facts.

Pre-registered held-out test. At R0 the test passes, but trivially: a model that convicts everything is predicted perfectly by a read-back that marks every element satisfied, and the gain (+33 points) equals the share of acquittal items. The informative rungs are R3 and R4, where the model applies part of the law. There the read-back still beats the statute in every cell (+0.5 to +22 points), but not by the pre-registered 0.15 in every framing. The test fails, and by our own rule these rungs report the per-element table (Figure 1), not the read-back.

model	lang	rung	read-back (Ja/Nein, verdict)	statute	gain	pre-registered verdict
70B	DE	R3	0.95, 0.98	0.81, 0.79	+14.8, +18.5 pts	fail
70B	DE	R4	0.88, 0.95	0.73, 0.74	+14.4, +21.8 pts	fail
70B	FR	R3	0.95, 0.96	0.83, 0.87	+12.0, +9.7 pts	fail
70B	FR	R4	0.87, 0.91	0.79, 0.70	+8.3, +21.3 pts	fail
70B	IT	R3	0.94, 0.96	0.79, 0.80	+14.8, +16.7 pts	fail
70B	IT	R4	0.92, 0.92	0.86, 0.74	+6.0, +18.5 pts	fail
8B Q4_K_M	DE	R3	0.94, 0.90	0.83, 0.79	+10.2, +11.1 pts	fail
8B Q4_K_M	DE	R4	0.78, 0.78	0.75, 0.77	+2.8, +1.4 pts	fail

Table 3. Held-out read-back test at the informative rungs (fit: wording families A, B; test: family C; R0 passes trivially for every model). Pass: gain ≥ 0.15 in every anchored framing.

Post-training and builds (exploration battery, 8B):

- At full precision, RLVR → DPO cuts correct acquittals under stated rules from 185 to 146 of 220 (R3, both framings) while keeping 104/104 convictions; weight blends move smoothly between the two.
- The public Q8_0 GGUF agrees with bf16 on 638 of 648 decisions, the Q4_K_M on 622 and our MLX 4-bit conversion on 548.

Dataset Limitations

- **Depth over breadth.** One statute modelled exactly, three of its five paragraphs; intent is fixed and the Abs. 4 dignity threshold is written to be met, which is where most court acquittals turn. The format is not tied to this statute: a second one (Art. 179ter) builds from five YAML files.
- **Exact twins need controlled wording.** The text is templated so that one fact changes at a time; the user-voice retelling shows the findings hold in everyday words.
- **Two legal choices a practitioner may contest, kept and stated.** The act is fixed as proven, so an Abs. 2 act ("propagating an ideology", which needs recruiting intent under BGE 140 IV 67) can occur in a one-off or private setting; and an online post from

before 1 July 2020 counts as completed on publication (dropping those 10 items per language moves no accuracy by more than 1.4 points, `scripts/sensitivity.py`). Acquittals are acquittals under Art. 261bis: a 1993/94 act is today also time-barred.

- **Every review is re-runnable.** Annotation, adjudication, court extraction and judging are agent runs with their instructions in the repository, so anyone can audit or repeat them; no private human judgment sits in the gold. Shared misconceptions of one model family are checked from outside it: the gold against the official text, mutation tests and 58 court decisions; the free-text judges by a judge of another model family (Qwen) and a deterministic parser.
- **Contested items are few** (15 per language). Two of the four contested settings (an e-mail to 25 colleagues, a club chat of 30) lean public after BGE 130 IV 111, so their criterion accepts a reasoned leaning; only a certain verdict without the publicity criteria fails.
- **Intervals** treat items as independent; the held-out gains carry twin-group bootstrap intervals (post hoc).

Lessons Learnt and Recommendations

For Apertus users.

- Do not ask about Art. 261bis without the law in context. With none, Apertus convicts every case we wrote, and its prose explains why.
- Do not count on retrieval: the pasted statute and leading case changed nothing. State the rule the case turns on in a plain sentence; the publicity rule is then applied. Still check characteristics against the closed list yourself, and state the date of the act.
- Prefer the Q8_0 GGUF or the served model to 4-bit MLX conversions.

For Apertus developers.

- DPO costs acquittals under stated rules (8B: 185 → 146 of 220 at R3): check preference pairs that reward condemning discrimination. The model obeys exclusions it is told but not "the list is closed", and it reads plain rules but not the law as written: instruction-following and long-context legal reading are training targets.
- Keep the `chosen` / `rejected` pairs out of training if you evaluate on them.

Reproducibility

`make run` (Docker, CPU, no key, no model) regenerates the dataset from the YAML (sha256 per file), recomputes every result from the stored responses, checks every request hash against its prompt, re-aggregates the judges, checks the pre-registration (`prereg-verify`: locked files, amendments A1–A11, the frozen held-out code) and the `claims.tsv` numbers, and exits non-zero on any mismatch. `LLM_*` variables or `make run-gguf` add a live slice against the published reference band. Seeds: generator 261, bootstrap 0 (2,000 resamples).

License

Code Apache-2.0; dataset CDLA-Permissive-2.0; this report CC-BY-4.0.

References

Art. 261bis and Art. 2 StGB, SR 311.0 (Fedlex, versions of 1 Jan 1995 and 1 Jul 2020); AS 1994 2887; AS 2020 1609. BGE 130 IV 111. Fan et al., LEXam, ICLR 2026. Han et al., LawShift, NeurIPS 2025 D&B. Servantez et al., OpenExempt, ACL Findings 2026. Huang et al., When Do LLMs Apply the Wrong Law?, arXiv 2608.14610. Röttger et al., Multilingual HateCheck, WOAH 2022.